

Exchange Concepts Knowledge Check — Variant 03

Purpose: verify that the student has read [How a Financial Exchange Works](#) and internalized the core principles of a modern exchange, across its full scope — market history, order mechanics, risk and compliance, and technology infrastructure.

Name/ID

Instructions

- There are 30 questions.
- Each question has five statements (A-E) which may be false or true.
- Select all statements you believe to be correct by clearly filling the square before the question.
- Each question may have multiple correct statements (for some, all five statements may be correct).
- There are always at least one correct statement.

Scoring

- +1 point for each correct option selected.
- -2 points for each incorrect option selected.
- 0 points for options not selected.
- Final test score is capped at a minimum of 0: if raw total is negative, recorded score is 0.
- A passing score is 70% of the maximum rounded down to the nearest integer.

Questions

1. Capital raising fundamentals

- ☐ A. Retained earnings are typically the fastest way to fund a time-sensitive growth opportunity
- ☐ B. A bond has a face value, a coupon rate, and a maturity date
- ☐ C. Shareholders have no guaranteed repayment obligation from the company
- ☐ D. In bankruptcy, equity holders are paid before bondholders
- ☐ E. Limited liability means a shareholder cannot lose more than their invested amount

2. IPOs, market cap, and share count

- ☐ A. Market capitalisation equals share price multiplied by shares outstanding
- ☐ B. A mega-IPO priced below \$200 a share necessarily indicates a smaller or lower-quality company
- ☐ C. Underwriters typically end up holding a large fraction of IPO shares they cannot sell
- ☐ D. New shares issued in an IPO deliver their sale proceeds directly to the company (or cashing-out early investors)
- ☐ E. A company's valuation and its share price are set independently of its chosen share count

3. Instruments beyond equities

- ☐ A. ETFs trade on exchanges in the same manner as individual stocks
- ☐ B. Futures and options are claims on assets that already exist
- ☐ C. Market indices can be traded directly on an exchange as a single security
- ☐ D. Index futures and index options allow investors to trade the performance of an index
- ☐ E. Bonds can trade over the counter as well as on exchange-based venues

4. The specialist and the physical book

- ☐ A. The NYSE specialist maintained a physical ledger of outstanding buy and sell orders
- ☐ B. The specialist's book had one column for bids and one for asks
- ☐ C. The word "book" in modern exchange software has no historical connection to a physical ledger
- ☐ D. Open outcry trading took place in a physical pit
- ☐ E. The stock ticker was invented in the 20th century

5. Wall Street and blue chips

- ☐ A. Wall Street is named after a wooden palisade built by Dutch colonists in 1653
- ☐ B. The term "blue chip" originates from casino poker chip denominations
- ☐ C. Oliver Gingold's 1923 usage described stocks trading at \$200 or more as "blue chip"
- ☐ D. A bear market is formally defined by a sustained rise of 20% or more from a trough
- ☐ E. "Going long" originated from a merchant's grain being held over time in a warehouse

6. Short selling operations

- ☐ A. The locate process is intended to confirm shares can be borrowed before a short sale is accepted
- ☐ B. Regulation SHO was adopted by the SEC in 2005
- ☐ C. A rising borrow rate can signal increasing short interest and limited supply
- ☐ D. Settlement for a short seller requires delivering shares they already own outright
- ☐ E. The exchange's matching engine distinguishes short sales from long sales at the point of matching

7. Real-world exchange facts

- ☐ A. Cboe is described as a major listed-options exchange
- ☐ B. HKEX and SGX are both mentioned as real-world exchanges in the document
- ☐ C. Nasdaq Stockholm is unrelated to the broader Nasdaq exchange group
- ☐ D. ASX refers to the Australian Securities Exchange
- ☐ E. Euronext is described in the document as a real-world exchange

8. Listing standards

- ☐ A. Initial listing standards commonly include a minimum bid price requirement
- ☐ B. Continued listing standards require no ongoing monitoring once a company is listed
- ☐ C. NASDAQ offers multiple listing tiers with differing thresholds
- ☐ D. A public float requirement ensures a genuine public market exists at listing
- ☐ E. Corporate governance requirements can include an independent audit committee

9. Limit order mechanics, numerically

- ☐ A. A buy limit order at \$150.40 crossing a \$150.30 ask fills at \$150.30
- ☐ B. In that scenario, the buyer receives \$0.10 per share of price improvement
- ☐ C. The buyer's limit price of \$150.40 represents the best price they could hope to receive
- ☐ D. A sell limit order's price acts as a floor, the minimum acceptable sale price
- ☐ E. Limit orders that cannot fill immediately are rejected rather than resting in the book

10. Market order risk

- ☐ A. Market orders offer a price guarantee but not an execution guarantee
- ☐ B. A market order submitted during a trading halt will not fill
- ☐ C. A market buy for 10,000 shares against only 100 available at the best ask can suffer significant slippage
- ☐ D. The 2010 Flash Crash involved a large sell algorithm selling into a thinning book
- ☐ E. Market orders can rest quietly in the book waiting for a better price

11. OCO, combo, and TIF review

- ☐ A. In a bracket order, only one of the take-profit or stop-loss legs will ultimately execute
- ☐ B. A GTC order can remain active across weekends and holidays
- ☐ C. A GTD order requires no scheduler tracking of an expiry date
- ☐ D. DAY is the default and most common time-in-force
- ☐ E. FOK orders are commonly used in arbitrage strategies requiring simultaneous full execution

12. Depth and imbalance

- ☐ A. Depth refers to the quantity resting at each price level
- ☐ B. Level 2 data is also called market depth data
- ☐ C. An imbalance ratio of exactly 0.5 indicates a heavily one-sided book
- ☐ D. Volume at touch refers only to the best bid and ask quantities
- ☐ E. A shallow market can be swept through multiple price levels by a single large order

13. Order book sweep, numerically

- ☐ A. Given resting asks of \$150.35/2,000, \$150.40/1,500, \$150.45/2,000, a 5,000-share buy sweeps all three levels
- ☐ B. The VWAP of that 5,000-share sweep is \$150.395
- ☐ C. The resulting market impact relative to the initial best ask is exactly \$0.05
- ☐ D. If a market sell for 3,500 hits bids of 2,000@\$150.34 and 1,500@\$150.33, both levels are fully exhausted
- ☐ E. After that sell sweep, the best bid moves to a lower price than \$150.33

14. Price-time priority scenarios

- ☐ A. A resting buy order amended from 1,000 to 500 shares keeps its original timestamp
- ☐ B. A resting buy order amended from \$150.30 to \$150.31 keeps its original timestamp
- ☐ C. Cancel-and-replace always produces a fresh timestamp at the back of the queue
- ☐ D. Under pro-rata with Order X (100 lots, arrived first) and Order Y (40 lots, arrived second) splitting 60 lots, Order X receives about 43 lots
- ☐ E. Under FIFO in that same scenario, Order Y would receive all 60 lots because it arrived after Order X's price was already set

15. Tick size specifics

- ☐ A. Storing \$150.30 as the integer 15030 avoids binary floating-point representation errors
- ☐ B. A submitted price that is not a valid tick multiple is accepted and later corrected by the matching engine
- ☐ C. CME's Treasury futures tick in fractions of 1/32nd of a point
- ☐ D. The EU's MiFID II tick size regime defines a single fixed tick for all price bands
- ☐ E. Stocks priced below \$1.00 in the US may be quoted in sub-penny increments

16. The matching engine and sweeps

- ☐ A. The sweep process works through price levels from best to worst until the order is filled or the book is exhausted
- ☐ B. NASDAQ's matching engines are described as acknowledging orders in under 100 microseconds in typical conditions
- ☐ C. CME Globex is described as handling on the order of 30-35 million messages per day across all products
- ☐ D. Multi-leg options strategies never require cross-symbol coordination
- ☐ E. Basket orders trading dozens of stocks as one instruction require cross-symbol scheduling

17. Order lifecycle and statuses

- ☐ A. NEW status means the order has been accepted and acknowledged
- ☐ B. A cancel message can move a PARTIAL order to CANCELLED
- ☐ C. An order can move directly from NEW to REJECTED
- ☐ D. FILLED orders remain resting in the book for potential future partial cancellation
- ☐ E. A GTD order that reaches its expiry date unfilled becomes EXPIRED

18. Market maker economics

- ☐ A. Market makers earn revenue primarily through the bid-ask spread
- ☐ B. A maximum spread obligation limits how wide a market maker's quote can be, measured in ticks
- ☐ C. A presence obligation might require quoting for at least roughly 85% of the session
- ☐ D. Adverse selection means a market maker systematically trades against better-informed counterparties
- ☐ E. Reduced fees and faster data access are examples of privileges granted in exchange for market maker obligations

19. Quote refresh and MMP recap

- ☐ A. Policy 3 (leave both sides active) is described as the riskiest refresh policy
- ☐ B. Eurex-style inactivation cancels both legs immediately upon any fill
- ☐ C. A typical Eurex MMP configuration might specify five fills within 500 milliseconds
- ☐ D. MMP is triggered manually by a human risk officer in real time
- ☐ E. Without MMP, market makers might quote wider spreads to compensate for adverse selection risk

20. Equilibrium price, a second worked scenario

- ☐ A. The equilibrium price maximises the minimum of cumulative bid and cumulative ask volume at that price
- ☐ B. If candidate price \$150 yields executable volume of 600 and candidate price \$151 yields 1,500, the algorithm should prefer \$151
- ☐ C. A seller who asked \$150 in an auction that clears at \$151 receives \$150, their original ask
- ☐ D. Remaining unfilled orders after an auction transition into the continuous book
- ☐ E. The uncross matches all eligible orders simultaneously at one single clearing price

21. Trading session details

- ☐ A. NYSE's continuous session runs from 9:30am to 4:00pm Eastern time
- ☐ B. Intraday auctions can be used as a softer alternative to a full circuit-breaker halt
- ☐ C. After-hours trading typically has the same liquidity and spreads as the regular session
- ☐ D. GTC orders reloaded at start-of-day are each re-acknowledged to the originating participant
- ☐ E. Book state saves provide the starting point for a warm restart if needed

22. Index weighting comparisons

- ☐ A. Equal-weighted indexes give every constituent the same influence regardless of size
- ☐ B. The DJIA is a market-capitalisation-weighted index
- ☐ C. A company that is 60% owned by its founding family contributes its full share count, unadjusted, in a free-float index
- ☐ D. Cap-weighted indexes give large companies more influence than small ones
- ☐ E. A rules-based index like Russell reconstitutes mechanically with less discretion than the S&P 500

23. Divisor math, a second scenario

- ☐ A. If the aggregate market cap of an index's constituents rises from \$7 trillion to a new value, the divisor is rescaled by the same ratio as the cap change
- ☐ B. This rescaling causes the index level to jump sharply at the moment of the corporate action
- ☐ C. A 2-for-1 stock split, on its own, leaves a company's market cap unchanged
- ☐ D. Ex-dividend price drops are typically absorbed via a divisor adjustment rather than shareholder value destruction
- ☐ E. The index level is computed as aggregate market cap divided by the divisor

24. Circuit breakers and collars

- ☐ A. LULD Tier 1 instruments generally have a narrower regular-hours band than Tier 2 instruments
- ☐ B. A 15-second monitoring period precedes a LULD trading pause once price exits its band
- ☐ C. Leveraged ETFs can have LULD bands multiplied by their leverage factor
- ☐ D. A static price collar is recalculated using the most recent trade rather than the prior close
- ☐ E. Market-wide circuit breakers did not apply before 9:30am during the March 2020 events

25. Clearing hierarchy and DvP

- ☐ A. Direct clearing members post margin directly to the CCP
- ☐ B. A non-clearing member such as a typical hedge fund routes trades through a clearing broker
- ☐ C. DvP eliminates the risk of one party delivering an asset while the other fails to deliver the corresponding side
- ☐ D. The clearing hierarchy has no bearing on who absorbs losses in a default
- ☐ E. US equities currently settle on a T+1 basis

26. VWAP and P&L, numerically

- ☐ A. Buying 100 shares at \$150 and 50 more at \$160 produces a VWAP of \$153.33
- ☐ B. If the market price then rises to \$160 while holding 150 shares at that VWAP, unrealised P&L is approximately \$1,000.50
- ☐ C. Realised P&L changes continuously as the market price moves, even without any further trade
- ☐ D. Unrealised P&L is a factor in triggering margin calls
- ☐ E. Selling all 150 shares at \$160 locks in the previously unrealised P&L as realised P&L

27. Clearly erroneous trades

- ☐ A. The clearly erroneous threshold generally widens as the reference price band gets lower
- ☐ B. A participant typically has an unlimited amount of time to file a clearly erroneous request
- ☐ C. Busting voids a trade entirely, reversing positions and cash movements
- ☐ D. Price adjustment can preserve that a trade occurred while correcting its price
- ☐ E. Multi-venue erroneous-price events require coordination so the same event is not busted on one exchange and left standing on another

28. Market abuse patterns

- ☐ A. Spoofing involves orders placed with no genuine intention to trade, cancelled once they've had their effect
- ☐ B. Front running involves a broker trading for their own account ahead of a known pending client order
- ☐ C. Quote stuffing is intended to consume matching-engine bandwidth and slow down competitors
- ☐ D. Insider trading is unrelated to the exchange's audit trail as evidence
- ☐ E. Navinder Singh Sarao's case established that spoofing can be prosecuted as manipulation without any single fraudulent order

29. Technology transport and protocols

- ☐ A. UDP multicast allows one packet to reach many subscribers simultaneously
- ☐ B. TCP is generally preferred for order submission because it guarantees reliable, in-order delivery
- ☐ C. A gap in a UDP sequence can be recovered via a separate TCP unicast channel
- ☐ D. ITCH messages require full text parsing at fixed offsets
- ☐ E. OUCH is described as the binary counterpart to FIX for order entry

30. Corporate actions and options mechanics

- ☐ A. A 4-for-1 stock split leaves total market capitalisation unchanged
 - ☐ B. Open limit orders must generally be adjusted for quantity and price after a stock split
 - ☐ C. American-style options can only be exercised on the expiry date itself
 - ☐ D. The OCC automatically exercises in-the-money options by as little as one cent absent a contrary instruction
 - ☐ E. Pin risk arises because a writer may not know until settlement whether they will be assigned
-